

Renormalization of the Shell-Model GT Operator from Chiral Effective Field Theory

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Reference: PRC **109**, 014301 (2024)

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Introduction: The Gamow-Teller Quenching Puzzle

- **The Puzzle:** Standard spherical shell model calculations systematically overestimate the Gamow-Teller (GT) strength. To match experimental decay rates, an empirical quenching factor $q \approx 0.75$ must be introduced to scale the axial-vector coupling g_A .
- **Possible Mechanisms:**

1. Operator Renormalization (Chiral Two-Body Currents)

Chiral Effective Field Theory (χ EFT) introduces two-body weak currents (TBC) which modify the transition operator and quench the effective g_A at the operator level.

2. Wavefunction Correlations (Many-Body Effects)

Multi-shell configuration mixing, intrinsic deformation, and complex many-body correlations.

Introduction: The Quenching Puzzle

- Using SRG-evolved NN and 3N interactions, the NCSM, the coupled-cluster method, and the VS-IMSRG are used to compute nuclei in different regions. Two-body currents contribute significantly to the quenching.

P. Gysbers et al., Nature Phys. 15, 428 (2019)

- Applying shell-model MBPT to construct an effective GT operator from χ EFT for pf -shell nuclei, the chiral expansion of the electroweak currents and the many-body renormalizations, share equal merit in providing a noteworthy reproduction of data.

L. Coraggio et al., Phys. Rev. C 109, 014301 (2024)

- Employing the projected shell model with chiral two-body currents for ^{76}Ge , many-body correlations (deformation, cross-shell) dominate GT quenching while two-body currents contribute only 5–15%, though the Hamiltonian uses a separable force not consistently derived from χ EFT.

H. Zhou, L.-J. Wang, Y. Sun, arXiv:2509.24542 (2025)

Introduction: The Quenching Puzzle

- Combining ab initio VS-IMSRG with chiral NN and 3N interactions and two-body currents for $N = 50$ neutron-rich isotones, The available experimental half-lives are well described once 2B currents are included. FF contributions suggests that they are smaller below $Z = 27$. Future work should include more detailed studies of the EFT interaction uncertainties and a consistent inclusion of FF contributions with 2B currents as well
[Z. Li, H. Miyagi, A. Schwenk et al., Phys. Rev. Lett. 136, 182501 \(2026\)](#)
- Using self-consistent RPA + PVC with chiral two-body currents for three doubly magic nuclei, ^{56}Ni , ^{100}Sn , and ^{132}Sn . PVC and two-body currents jointly yield $q \approx 0.73\text{--}0.80$ and reproduce experimental GT strengths. Among the several contributions of chiral TBC, the meson-exchange terms give the major contribution to the quenching, to the choice of LEC parameters, while the momentum-dependent term has a negligible contribution. [B.-L. Wang et al., arXiv:2605.21218 \(2026\)](#)

Previous Work and Motivation

Previous approach (Coraggio et al. 2017–2022):

- Derived effective SM GT operators via MBPT
- Starting from CD-Bonn NN potential + $V_{\text{low-k}}$ renormalization
- **No** two-body electroweak currents included
- Still reproduced GT data without explicit quenching

This work

- Consistent derivation of both H_{eff} and from **chiral EFT** (same LECs for forces and currents)
- Renormalization:
 - Two-body electroweak currents (χEFT expansion up to $N^3\text{LO}$)
 - Many-body correlations (MBPT effective operator): Shell model retains only a finite model space P ; configurations in $Q = 1 - P$ must be accounted for by **effective operators**.
- Goal: **separate** the relative contributions of 2BC vs. model-space truncation



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Chiral Nuclear Hamiltonian

Starting Hamiltonian from χ EFT:

$$H = H_{2\text{NF}}^{\text{N}^3\text{LO}} + H_{3\text{NF}}^{\text{N}^2\text{LO}}$$

- **2NF** at N^3LO : Entem–Machleidt potential, $\Lambda = 500$ MeV
- **3NF** at N^2LO :
 - 2π exchange $\rightarrow$ LECs c_1, c_3, c_4 (shared with 2NF)
 - 1π exchange $\rightarrow$ new LEC c_D
 - Contact $\rightarrow$ new LEC c_E
- $c_D = -1, c_E = -0.34$ (fixed from $A = 3$ observables, Navrátil et al. 2007)

Forces and currents share the same LECs $\Rightarrow$ fully consistent framework.

Chiral Axial Current: One-Body Part

The axial current is expanded up to N³LO:

$$J_A = J_A(1b) + J_A(2b)$$

LO (one-body):

$$J_{A\pm}^{\text{LO}} = -g_A \sum_i \boldsymbol{\sigma}_i \tau_{i,\pm} \quad (1)$$

This is the standard GT operator with $g_A = 1.2723$.

N²LO (one-body, relativistic correction:

$$J_{A\pm}^{\text{N}^2\text{LO}} = \frac{g_A}{2m_N^2} \sum_i \mathbf{K}_i \times (\boldsymbol{\sigma}_i \times \mathbf{K}_i) \tau_{i,\pm} \quad (2)$$

where $\mathbf{K}_i = (\mathbf{p}'_i + \mathbf{p}_i)/2$ is the average momentum.

This correction is typically small ($\sim \mathcal{O}(p^2/m_N^2)$) but included for consistency.

Chiral Axial Current: Two-Body Part (1π Exchange)

$N^3\text{LO } 1\pi\text{-exchange}$:

$$J_{A\pm}^{N^3\text{LO}}(1\text{PE}; \mathbf{k}) = \sum_{i < j} \frac{g_A}{2f_\pi^2} \left\{ 4c_3 \tau_{j,\pm} \mathbf{k} + (\boldsymbol{\tau}_i \times \boldsymbol{\tau}_j)_\pm \right. \quad (3)$$
$$\times \left[\left(c_4 + \frac{1}{4m} \right) \boldsymbol{\sigma}_i \times \mathbf{k} - \frac{i}{2m} \mathbf{K}_i \right] \left. \right\} \boldsymbol{\sigma}_j \cdot \mathbf{k} \frac{1}{\omega_k^2} + (i \rightleftharpoons j)$$

- $\omega_k^2 = k^2 + m_\pi^2$
- $\mathbf{k} = (\mathbf{p}'_i - \mathbf{p}_i)/2$ (momentum transfer)
- Contains c_3, c_4 — same LECs as in the chiral 3NF 2π -exchange!

Chiral Axial Current: Two-Body Part (Contact Term)

N³LO contact term :

$$J_{A\pm}^{\text{N}^3\text{LO}}(\text{CT}; \mathbf{k}) = \sum_{i < j} z_0 (\boldsymbol{\tau}_i \times \boldsymbol{\tau}_j)_{\pm} (\boldsymbol{\sigma}_i \times \boldsymbol{\sigma}_j) \quad (4)$$

The coupling constant z_0 is **not free**:

$$z_0 = \frac{g_A}{2f_{\pi}^2 m_N} \left[-\frac{m_N}{4g_A \Lambda_{\chi}} c_D + \frac{m_N}{3} (c_3 + 2c_4) + \frac{1}{6} \right] \quad (5)$$

- c_D — same LEC as 3NF 1π exchange
- $c_3 + 2c_4$ — same LECs as in 3NF and 1PE current
- $1/6$ — fixed by chiral symmetry (no free parameter)
- No new LECs are introduced at N³LO!

From Momentum Space to Configuration Space

The two-body current in configuration space:

$$J_{A\pm}(2b) = \int \frac{d\mathbf{k}}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}_{ij}} C_{\Lambda}(k) J_{A\pm}(2b; \mathbf{k}) \quad (6)$$

- $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$
- Regulator: $C_{\Lambda}(k) = e^{-(k/\Lambda)^4}$ with $\Lambda = 500$ MeV (same cutoff as the NN potential)
- $\mathbf{K}_j$ is replaced by $-i\nabla_j$ when transforming to r -space
- Matrix elements calculated in the **HO basis**, consistently with the chiral Hamiltonian

The 2BC is included explicitly (full two-body matrix elements), not via normal-ordering approximation as in Menéndez et al. (2011).

Effective Shell-Model Hamiltonian

Goal: Project the full Hamiltonian onto a truncated model space P .

$\hat{Q}$ -box folded-diagram expansion (Kuo–Osnes framework)

$$\hat{Q}(\epsilon) = PH_1P + PH_1Q \frac{1}{\epsilon - QHQ} QH_1P \quad (7)$$

expanded perturbatively:

$$\frac{1}{\epsilon - QHQ} = \sum_{n=0}^{\infty} \frac{1}{\epsilon - QH_0Q} \left(\frac{QH_1Q}{\epsilon - QH_0Q} \right)^n \quad (8)$$

Truncation:

- NN vertices: up to 3rd order in perturbation theory
- NNN vertices: up to 1st order (normal-ordered 3BF)
- Density-dependent for each nuclear system (accounts for induced 3BF among valence nucleons)

Effective Shell-Model Transition Operators

The effective operator is derived via the same MBPT framework.

Define $\hat{\Theta}$ -box vertex functions:

$$\chi_0 = (\hat{\Theta}_0 + \text{h.c.}) + \hat{\Theta}_{00} \quad (9)$$

$$\chi_1 = (\hat{\Theta}_1 \hat{Q} + \text{h.c.}) + (\hat{\Theta}_{01} \hat{Q} + \text{h.c.}) \quad (10)$$

$$\chi_2 = (\hat{\Theta}_1 \hat{Q}_1 \hat{Q} + \text{h.c.}) + (\hat{\Theta}_2 \hat{Q} \hat{Q} + \text{h.c.}) + \dots \quad (11)$$

Then:

$$= \hat{Q}^{-1} (\chi_0 + \chi_1 + \chi_2 + \dots) \quad (12)$$

Truncation:

- χ_n series arrested at $n = 2$
- One-body component: up to 3rd order in MBPT
- Two-body component (from 2BC): up to 2nd order
- The two-body current also contributes a normal-ordered one-body term (evaluated with respect to the closed-core reference state)

GT and $2\nu\beta\beta$ Matrix Elements

Single GT transition:

$$M_{\text{GT}}^{1\nu} = \langle J_f || \mathcal{J}_A^{\mathcal{I}} || J_i \rangle \quad (13)$$

$2\nu\beta\beta$ decay:

$$M_{\text{GT}}^{2\nu} = \sum_n \frac{\langle 0_f^+ || \mathcal{J}_A^{\mathcal{I}} || 1_n^+ \rangle \langle 1_n^+ || \mathcal{J}_A^{\mathcal{I}} || 0_i^+ \rangle}{E_n + E_0} \quad (14)$$

- $\mathcal{I}$: bare or effective operator
- $E_0 = \frac{1}{2}Q_{\beta\beta}(0^+) + \Delta M$

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Four Calculations to Disentangle the Two Effects

To assess the **separate** roles of 2BC and model-space truncation:

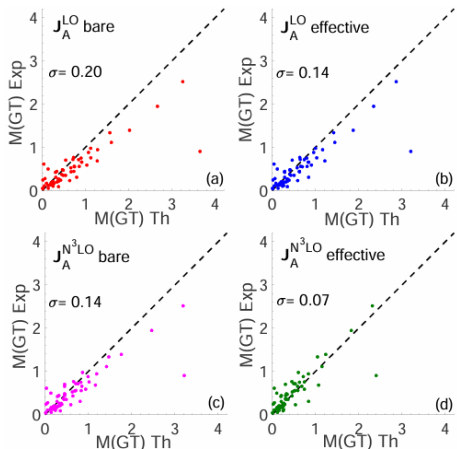


Figure 1: Correlation plots between experimental and calculated values of the GT NME

- The values of σ obtained with the operators (b) and (c) show that for the $0f1p$ shell the improvements we obtain with respect to the results with the bare operator (a) can be equally ascribed to the renormalization of the GT operator by way of the ChPT expansion of JA
- As well as to the derivation of SM effective operators which account for the configurations outside the model space

^{48}Ca : GT Strength Distribution

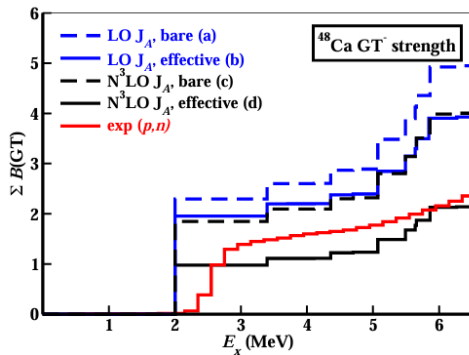
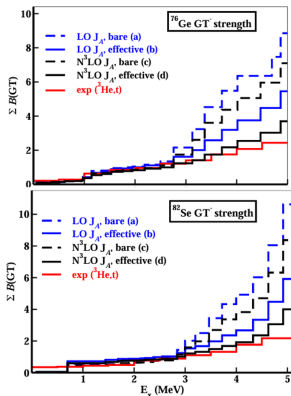


Figure 2: Running sums of the ^{48}Ca B(GT) strengths as a function of the excitation energy E_x

- Bare operator (a) **overestimates** the total GT strength
- Effective LO operator (b) reduces strength by $\sim 30\%$
- Bare $N^3\text{LO}$ operator (c) provides similar reduction
- Full operator (d): best agreement with data from charge-exchange reactions

^{76}Ge and ^{82}Se : GT Strength

For $0f_{5/2}1p0g_{9/2}$ shell ($0\nu\beta\beta$ candidates):



- Model space $0f_{5/2}1p0g_{9/2}$ lacks spin-orbit partners $\Rightarrow$ larger renormalization needed
- Operator (d) gives a dramatic reduction of GT strength ($\sim 50\%$ from bare)
- Both 2BC and effective operator are essential for heavy nuclei

Figure 3: Running sums of the ^{76}Ge and ^{82}Sn $B(\text{GT})$ strengths as a function of the excitation energy

$2\nu\beta\beta$ Nuclear Matrix Elements

Table III. Experimental [90, 91] and calculated $M^{2\nu}$'s (in MeV^{-1}) for ^{76}Ge and ^{82}Se $2\nu\beta\beta$ decay.

$^{76}\text{Ge} \rightarrow ^{76}\text{Se}$						
$J_i^\pi \rightarrow J_f^\pi$	(a)	(b)	(c)	(d)	Expt	
$0_1^+ \rightarrow 0_1^+$	0.211	0.153	0.160	0.118	0.129 ± 0.004	
$0_1^+ \rightarrow 2_1^+$	0.023	0.042	0.025	0.048	≤ 0.035	
$0_1^+ \rightarrow 0_2^+$	0.009	0.086	0.016	0.063	≤ 0.089	
$^{82}\text{Se} \rightarrow ^{82}\text{Kr}$						
$J_i^\pi \rightarrow J_f^\pi$	(a)	(b)	(c)	(d)	Expt	
$0_1^+ \rightarrow 0_1^+$	0.173	0.123	0.136	0.095	0.103 ± 0.001	
$0_1^+ \rightarrow 2_1^+$	0.003	0.006	0.008	0.033	≤ 0.020	
$0_1^+ \rightarrow 0_2^+$	0.018	0.007	0.013	0.007	≤ 0.052	

- Operator (d) reproduces experiment **very well**
- Results comparable to previous CD-Bonn calculations
- For $0_1^+ \rightarrow 2_1^+, 0_2^+$ transitions: intermediate-state contributions **cancel**, giving small $M^{2\nu}$ (consistent with upper bounds)

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Summary

- 1 First consistent treatment of nuclear Hamiltonian and electroweak currents within MBPT for shell-model calculations, starting from χ EFT.
- 2 Chiral two-body electroweak currents (N^3 LO) and Model-space truncation (effective operator via MBPT) are needed to reproduce experiment.
- 3 $2\nu\beta\beta$ NME Results validate the approach for future $0\nu\beta\beta$ NME calculations (^{76}Ge , ^{82}Se , ^{100}Mo , ...).